MD-88 Audio Companion

Bring-up report: channels, findings, fixes and next steps

Boards: production order Y15 (5 built) and KiCad project v19_r3. Display: Elecrow CrowPanel Advance 4.3 inch (ESP32-S3). Diagnostics firmware: branch claude/sharp-wright-0b4a35, commit cf3875b. Report date: 12 November 2026.

1. Where things stand

Area	Status	Notes
Display, touch, Wi-Fi update	Working	LVGL diagnostics screen; firmware updates over the AudioBox-Update hotspot.
I2C devices	Working	Codec, expander and haptic driver all answer on every board tested.
Headphone output	Working with a bench bridge	1 kHz tone heard in both ears on PHONES and on the Lemo, after bridging the HP relay.
Bypass relays	Design bug	Relay coils are wired with reversed polarity, so neither relay ever switches.
Microphone input	Working with radio-supplied bias	Voice passes mic to headset through the board when an aviation handheld supplies mic bias.
On-board mic bias	Too weak	The 3.3 V bias through 2.2 kohm does not run the aviation mic and carries supply noise.
12 V boost (ANR power)	Not tested	Needs the battery; the boost runs only from the battery connector.
Encoder, haptic motor	Not tested	Firmware is written; parts were on order.

2. The channels

The codec is an ES8388 running as an I2S slave at 48 kHz, 16-bit stereo. The ESP32 supplies all clocks.

2.1 Audio outputs

Channel	Path	Where it appears
Headphone pair (L, R)	Codec pins 15 / 14, 10 uF, TPA6111A2 amp (gain -1), 220 uF, 10 ohm, HP relay contacts 4 and 5 (closed only when the relay is energized)	PHONES (L R G) and the Lemo header L and R pads. These share one net.
Line pair (L, R)	Codec pins 12 / 11 through coupling capacitors, no amplifier	LINE OUT and OUT 2 (same signal, line level, too quiet for a headset).
Aircraft audio (L, R)	AIRCRAFT connector L and R, HP relay contacts 2 and 7 (closed when the relay is off)	PHONES and Lemo whenever the relay is off. This is the fail-safe pass-through.

2.2 Audio inputs

Channel	Path	Status
Microphone	MIC IN pin M and Lemo pad M (same net, MIC_RAW), 1 uF, MCP6022 preamp (gain 23, about 27 dB), 1 uF, codec pin 24	Confirmed on codec input 1, left channel. The right channel of input 1 is grounded and unused.
LINE IN (L, R)	LINE IN connector through coupling capacitors to codec pins 22 / 21	Expected on codec input 2. Not yet confirmed; the loopback test still selects input 1 and needs re-checking.

Channel	Path	Status
Mic to aircraft radio	Relay off: raw mic straight to AIRCRAFT pin M. Relay on: preamp output through 1 uF to AIRCRAFT pin M.	Raw path confirmed with the handheld. The processed path cannot work until the relay coil is fixed.

2.3 Connectors

Silkscreen	Pins	Purpose
PHONES	L, R, G	Headset headphone jack. Amp output or aircraft audio, chosen by the HP relay.
LEMO header	L, G, R, M, G, +	Headset on a 6-pin Lemo. Bose wiring: L = pin 3, G = pin 2, R = pin 4, M = pin 5, G = pin 6, + = pin 1 (12 V).
MIC IN	M, G	Headset mic. M must reach the RING of the mic plug, so the jack needs three contacts.
AIRCRAFT	L, R, G, M, P	To the aircraft or handheld: headphone audio in, mic out, push-to-talk pass-through.
LINE IN / LINE OUT / OUT 2	L, R, G	Line-level audio in and out.
PTT	P, G	Push-to-talk pass-through to the AIRCRAFT connector.
ATC BTN and CP J13	see note	Button input BTN_IN. Currently unreadable (see finding 6).
ENCODER	8 pins	Dual concentric encoder with push, read through the PCA9534 expander.
PANEL	6 pins	Power button, relay gate, spare input, 3.3 V.
HAPTIC	+, -	Vibration motor from the DRV2605L.
BATT	+, -	Single-cell LiPo. Feeds only the 12 V boost.

2.4 Control signals and LEDs

Signal	ESP32 pin	Notes
I2S MCLK / BCLK / LRCK	IO19 / IO5 / IO6	MCLK 12.288 MHz measured on the scope.
I2S data out / in	IO4 / IO2	IO4 feeds the codec DAC; IO2 receives the codec ADC.
I2C SDA / SCL	IO15 / IO16	Shared with the Elecrow touch and backlight chips. Codec 0x10, expander 0x38, haptic 0x5A.
ANR_POWER_EN	IO8	Turns on the 12 V boost (about 12.6 V by the feedback divider).
BYPASS_EN	IO20	High energizes the relays, which selects the AMP and processed mic. The name is misleading: high means ACTIVE, low means bypass.

LED	Colour	Meaning
PWR	Green	3.3 V present.
ACT	Yellow	IO20 is high: active mode requested.
ANR	White	12 V rail is up (needs the battery).
BYP	Red	Relay driver is off: bypass (pass-through) mode.

3. What we learned

1. Relay coil polarity is reversed (confirmed, all boards and v19_r3)

The Omron G6K-2F-Y datasheet shows coil pin 1 as + and pin 8 as -. On the board, pin 1 goes to the switching transistor (ground side) and pin 8 goes to +3.3 V. This relay is polarity-sensitive, so it never operates. PHONES and Lemo therefore always received the aircraft path and never the amp. This was the main cause of the silence. The contact assignment is otherwise correct: 2-3 and 7-6 closed at rest, 3-4 and 6-5 closed when energized.

2. On-board mic bias is too weak and noisy (strongly indicated)

The BIAS jumper feeds 3.3 V through 2.2 kohm. The mic then draws well under 1 mA and produces no usable audio, and Wi-Fi ripple on the 3.3 V supply is amplified into static and pulsing. With an aviation handheld supplying bias, the voice came through the board cleanly enough to hear. A direct test with a proper bias source on the bench has not been done yet.

3. Codec recording volume defaults to -96 dB (confirmed, fixed in firmware)

Registers 0x10 and 0x11 read 0xC0 after power-up. The firmware now writes 0 dB at every start.

4. Codec powers up as clock master (confirmed, fixed in firmware)

The ES8388 drove BCLK and LRCK and fought the ESP32, which held IO5 and IO6 high. The firmware now sets slave mode at boot.

5. IO19 and IO20 are the ESP32-S3 USB pins (fixed in firmware)

The USB block can override those pads. The firmware now clears the override before driving IO20. The self-test checks this.

6. BTN_IN lands on the expander INT pin (confirmed design bug)

U3 pin 13 is INT, an output. The button on ATC BTN / CP J13 cannot be read as built.

7. The mic jack must have three contacts

On an aviation mic plug the mic is on the ring and the tip is push-to-talk. A two-contact jack reaches only tip and sleeve.

8. JLCPCB did not fit the through-hole connectors

PHONES, LINE IN/OUT, MIC IN, AIRCRAFT, PTT, HAPTIC, BATT and the headers were hand-soldered. They are not in the assembly BOM.

9. One Elecrow display board is faulty

It does not drive IO20, so the bypass LEDs never changed on any companion board paired with it. Another has a stuck BOOT button. Keep the faulty one out of tests.

10. Bench notes

Keep boards off conductive surfaces; one shorted on the table. The Klein MM600 does not read small 1 kHz audio reliably, so use the on-screen Mic level bar and the self-test instead. Companion board B showed an intermittent codec connection on one occasion and should be re-tested.

4. The fixes

4.1 Rework for the boards already built

Fix	How	Status
Relay coil (both relays)	Lift the legs of pins 1 and 8 off their pads and cross them with two short wires: pin 1 leg to the pin 8 pad (+3.3 V), pin 8 leg to the pin 1 pad (transistor). Then remove the bench solder bridge on the HP relay.	Proposed. Not yet tried on a board.
Bench bridge (temporary)	HP relay only: bridge pins 3-4 and pins 5-6. Amp is then always on PHONES and Lemo. Never bridge the MIC relay. Do not fly with it; it removes the fail-safe.	Done on two boards. Works.
Mic bias	Remove the BIAS solder bridge. Feed the BIAS pad on the 2.2 kohm side from the 12 V rail (test point TP5 or the Lemo + pad) through an RC filter, and reduce the series resistor toward the usual aviation value (about 470 ohm). Needs the battery and ANR power on.	Proposed. Values to be validated on the bench first.
Button input	Cut the BTN_IN trace at U3 pin 13 and jumper it to pin 12 (SPARE5); update the firmware map.	Proposed.
Mic jack	Three-contact aviation mic jack: ring lug to M, barrel lug to G, tip unconnected.	Part to buy.

4.2 Changes for the next board revision (KiCad)

- Swap the two coil nets on HEADPHONE_BYPASS-1 and MIC_BYPASS-1: pin 1 to +3V3, pin 8 to K1_COIL_DRIVE. Add + and - to the relay symbol pins so the polarity is visible.
- Replace the 3.3 V mic bias with a filtered bias from the 12 V rail, with a series resistor suited to aviation mics.
- Move BTN_IN from U3 pin 13 (INT) to a real input pin such as pin 12.
- Review the MIC relay spare contacts: in v19_r3 pins 5, 6, 7 are wired in parallel with the right headphone channel. Confirm this is intended.
- Keep IO19 / IO20 in mind: they are the ESP32-S3 USB pins. Moving MCLK and BYPASS_EN to other pins would keep USB debugging available.
- Rename BYPASS_EN to something like ACTIVE_EN, since high means active.
- Update the title block notes (they still say ES8311 and three JST cables) and add the hand-fitted connectors to the assembly BOM if JLCPCB should fit them.

4.3 Firmware

- Done: slave mode at boot, recording volume 0 dB, USB pad release for IO20, mic on codec input 1, digital sidetone to both ears, self-test, mic scan, loopback test, web bench tools.
- Reverb in the sidetone: the mic is copied through the ESP32 in blocks, which adds a short delay, and the handheld adds its own sidetone on top. Fix by shrinking the I2S buffers or by using the codec's analog mixer for sidetone, and by using only one sidetone source.
- Rename the on-screen Bypass switch to Active, re-check which codec input LINE IN uses, and bring back the Haptic button.

5. Next steps, in order

#	Step	What it proves
1	Connect the battery (check polarity first: pin 1 is +), switch ANR power on, confirm the white ANR LED and about 12 V on TP5.	The 12 V boost works.

#	Step	What it proves
2	Bench-test the mic with a proper bias (9 V battery through about 1 kohm onto M, or the 12 V rail) and no radio attached.	The mic bias fix, before any rework.
3	Do the relay coil rework on one board. Remove the HP bridge. Flip the switch and listen for the relay click.	Bypass switching works as designed.
4	With the relay fixed: switch off = aircraft audio on the headset; switch on = board audio.	The fail-safe and active paths.
5	Rework the mic bias on that board and repeat the mic monitor test without the handheld.	The board can power the mic itself.
6	Fit the three-contact mic jack; test encoder and haptic motor when they arrive.	Remaining inputs and outputs.
7	Fix the sidetone delay in firmware, then start the real application firmware on top of the diagnostics.	Usable audio quality.
8	Apply section 4.2 in KiCad and order the next revision.	A board that needs no rework.

6. Bench tools reference

Tool	How to use it
Firmware update	Tap Firmware update, join Wi-Fi AudioBox-Update (password audiobox1), open http://192.168.4.1 and upload build/diagnostics.bin.
Self-test report	http://192.168.4.1/report (also the Self-test button). The loopback line needs LINE OUT jumpered to LINE IN.
Mic scan	http://192.168.4.1/micscan while talking for about 10 seconds. Reports each codec input and selects the mic input.
Register access	http://192.168.4.1/i2c?dev=0x10&reg=0x10 reads; add &val=0x00 to write. Values are lost at power-off.
Tone / pins	http://192.168.4.1/tone?on=1 and /gpio?pin=5&mode=out&level=0 (pins 4, 5, 6, 19 only).
Build on the Mac	cd ~/AudioBox/idf_diagnostics, then git pull and idf.py build (after sourcing the ESP-IDF activate script).

Sources: production gerbers (Y15), KiCad v19_r3 PCB and schematic, the Omron G6K datasheet supplied during testing, the Bose panel connector instructions, and bench results reported during the session. Items marked proposed have not been tried on hardware.